Influenza in Children

134

Introduction

- ☑ Influenza is an acute viral respiratory illness caused by influenza A or B viruses.

Epidemiology

- ☑ *Agent*: The influenza virus is single-stranded ribonucleic acid (RNA) virus belonging to *orthomyxoviridae* family and is classified into three types A, B, and C. The virus is pleomorphic spherical or filamentous in shape. 80–120 nm in size, single-stranded eight segmented RNA virus. There are 16 subtypes of hemagglutinin surface proteins and nine serotypes of neuraminidase. The human influenza virus subtypes contain mainly H1, H3, and rarely H2 (does not exist in circulation), and N1 and N2 surface proteins.
- ☑ *Host*: It is a zoonotic virus causing infection in animals such as pigs, horses, and seals, birds such as fowls, ducks, and chicken, and humans, reassorting with each other.
- ☑ *Transmission*: Droplet infection and fomites.
- ☑ *Incubation period:* 1–7 days.
- ☑ *Communicability*: From 1 day before to 7 days after the onset of symptoms. Children may spread the virus for a longer period.

Case Definitions

Used in influenza pandemic.

- ☑ *A suspected case* of influenza virus infection is defined as a person with acute febrile respiratory illness (fever ≥38°C) with onset:
  - Within 7 days of close contact with a person who is a confirmed case of influenza virus infection, or within 7 days of travel to community where there are one or more confirmed influenza cases, or
  - Resides in a community where there are one or more confirmed influenza cases.
- ☑ *A probable case* of influenza virus infection is defined as a person with an acute febrile respiratory illness who:
  - Is positive for influenza, but unsubtypable for H1 and H3 by influenza reverse transcription polymerase chain reaction (RT-PCR) or reagents used to detect seasonal influenza virus infection, or
  - Is positive for influenza by an influenza rapid test or an influenza immunofluorescence assay (IFA) plus meets criteria for a suspected case.
  - Individual with a clinically compatible illness who died of an unexplained acute respiratory illness who is considered to be epidemiologically linked to a probable or confirmed case.
- ☑ *A confirmed case* of influenza virus infection is defined as a person with an acute febrile respiratory illness with laboratory confirmed virus infection at world health organization (WHO) approved laboratories by one or more of the following tests:
  - Real time PCR
  - Viral culture
  - Four-fold rise in influenza virus specific neutralizing antibodies.

Diagnosis

- ☑ Routine investigations required include hematological, biochemical, radiological, and microbiological tests as necessary.
- ☑ For confirmation of diagnosis, clinical specimens such as nasopharyngeal swab, throat swab, nasal swab, wash or aspirate, and tracheal aspirate (for intubated patients) are to be collected by a trained microbiologist preferably before administration of the antiviral drug.
- ☑ Specimens should be either transported at 4°C in viral transport media for testing within 24 hours or need to be stored at –70°C. Confirmatory tests include:
- ☑ Real time RT-PCR.
- ☑ Isolation of the virus in culture.
- ☑ Four-fold rise in virus specific neutralizing antibodies.

Clinical Picture

Influenza is an acute febrile highly contagious respiratory infection characterized with sudden onset of fever (≥38°C), chills, myalgia, headache, fatigue, usually followed by pharyngitis, rhinitis, nonproductive cough, and sometimes cervical lymphadenopathy. In young infants, it may present as gastroenteritis and infrequently like croup or bronchiolitis. The recovery is usually uneventful in most of the cases within 7–8 days, however, influenza can cause severe illnesses and complications leading to hospitalization and deaths mainly in the high risk groups such as very young, elderly, pregnant women, and subjects having chronic illnesses as well as immunosuppression.

Management

Category A	***Category B***	***Category C***
☑ Patients with mild fever plus cough/sore throat with or without body ache, headache, diarrhea, and vomiting will be categorized as Category-A ☑ They do not require oseltamivir and should be treated for the symptoms with bed rest, plenty of fluids, and paracetamol for fever ☑ The patients should be monitored for their progress and reassessed at 24–48 hours by the doctor with close monitoring of the vitals and appearance of any red flags ☑ No testing of the patient for H1N1 is required. Most of the time disease is not severe enough to warrant antivirals and is self-limiting ☑ Patients should confine themselves at home and avoid mixing up with public and high risk members in the family. Usually no laboratory testing is required for such patients	☑ In addition to all the signs and symptoms mentioned under Category-A, if the patient has high grade fever and severe sore throat, requires home isolation and oseltamivir ☑ In addition to all the signs and symptoms mentioned under Category-A, individuals having one or more of the following high risk conditions shall also be treated with oseltamivir ☑ Children with mild illness but with predisposing risk factors ☑ Pregnant women ☑ Persons aged 65 years or older ☑ Patients with lung diseases, heart disease, liver disease, kidney disease, blood disorders, diabetes, neurological disorders, cancer, and HIV/AIDS ☑ Patients on long-term cortisone therapy ☑ Laboratory investigations such as rapid influenza diagnostic tests (RIDTs), RT-PCR may be required in these cases. Antivirals should be started soon, preferably within 48 hours	☑ In addition to the above signs and symptoms of Category-A and B, if the patient has one or more of the following: breathlessness, chest pain, drowsiness, fall in blood pressure, sputum mixed with blood, bluish discoloration of nails ☑ Children with influenza such as illness who had a severe disease as manifested by the red flag signs (somnolence, high and persistent fever, inability to feed well, convulsions, shortness of breath, difficulty in breathing, etc) ☑ Worsening of underlying chronic conditions ☑ All these patients aforementioned in Category-C require testing, immediate hospitalization, and treatment

Drugs

There are two groups of drugs
1. NA inhibitor enzymes such as oseltamivir and zanamivir.
2. M2 protein inhibitors such as amantadine and rimantadine.

Oseltamivir is licensed for chemoprophylaxis and is used in children above 3 months of age **(Table 1)**. Zanamivir is approved in children above 7 years of age only. There is no role of antibiotics in primary influenza infections, however, sometimes when they are complicated with the superadded bacterial infections, in such situations where the clinical picture is very strongly suggestive of coexisting bacterial infection, the clinician should start appropriate antibiotics concomitantly.

Table 1: Oseltamivir is the recommended drug both for chemoprophylaxis and treatment.

By age/ weight	***Dose for prophylaxis***	***Dosing for prophylaxis***	***Duration for prophylaxis***	***Dose for treatment***	***Dosing for treatment***	***Duration for treatment***
For infants						
<3 months	Use of oseltamivir for chemoprophylaxis is not recommended unless situation is judged critical due to limited data in this age group	NA	NA	12 mg/dose	BD	5 days
3–5 months	3 mg/kg/dose	OD	7 days	20 mg/dose	BD	5 days
6–11 months				25 mg/dose		
For children >1 year						
<15 kg	30 mg/dose	OD	7 days	30 mg/dose	BD	5 days
15–23 kg	45 mg/dose	OD	7 days	45 mg/dose	BD	5 days
24–40 kg	60 mg/dose	OD	7 days	60 mg/dose	BD	5 days
>40 kg	75 mg/dose	OD	7 days	75 mg/dose	BD	5 days

If needed, dose and duration can be modified as per clinical condition.

Adverse reactions: Oseltamivir is generally well-tolerated. Gastrointestinal side effects (transient nausea, vomiting) may increase with increasing doses, particularly above 300 mg/day. Occasionally, it may cause bronchitis, insomnia, vertigo, abdominal pain, epistaxis, bronchitis, otitis media, dermatitis, and conjunctivitis.

Prevention

Vaccination is the only and the most effective modality to prevent influenza transmission and to control epidemics and pandemics.

Influenza vaccines are of two categories:

1. *Live attenuated*: Trivalent
2. *Inactivated influenza*: Trivalent and quadrivalent

Recently introduced quadrivalent vaccine has two strains of A H1N1 and H3N2, and two strains of B lineages (commonly affecting children) namely Victoria and Yamagata.

IAP Recommendations for Vaccination

- ☑ It is recommended in children of 6 months to 5 years of age and those with high risk conditions.
- ☑ First time vaccinee between 6 months and 8 years should receive two doses 1 month apart; while for those above 9 years of age a single dose is sufficient.
- ☑ Annual revaccination is with a single dose.
- ☑ Dosage of inactivated influenza vaccines (IIVs) is 0.5 mL IM.
- ☑ All the currently available IIVs in the country contain the swine flu or A H1N1 antigens, hence no need to vaccinate separately.

Further Reading

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Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Hypospadias

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